

NIVEDITA SEN

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ABOUT

Motivated and enthusiastic B.Tech CSE student at UEM Kolkata with a strong interest in Artificial Intelligence, Machine Learning, IoT, and Software Development. Experienced in developing projects involving AI-powered systems, smart automation, and data analysis, with hands-on knowledge of Python, Java, C, and IoT technologies. Passionate about problem-solving, Data Structures & Algorithms (DSA), and building innovative solutions to real-world challenges.

EDUCATION

Institute of Engineering & Management, Newtown (UEM Kolkata) **2028**
B.Tech – CSE (IoT, Cyber Security & Blockchain Technology) *Kolkata, West Bengal*

DAV Koylanagar – CBSE Board **2023**
Senior Secondary Examination (Class XII) *Dhanbad, Jharkhand*

Carmel School – ICSE Board **2021**
Secondary Examination (Class X) *Dhanbad, Jharkhand*

RELEVANT COURSEWORK

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|--------------------------------|-------------------------------|---------------------|
| • Data Structures & Algorithms | • Object-Oriented Programming | • Machine Learning |
| • Database Management Systems | • Operating Systems | • Computer Networks |

TECHNICAL SKILLS

Languages: C, Python, Java

Web Technologies: HTML, CSS, JavaScript

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming

Machine Learning: Data Preprocessing, Model Training & Evaluation, Scikit-Learn, TensorFlow

Tools & Platforms: Git, GitHub, VS Code

Soft Skills: Problem-Solving, Teamwork, Adaptability, Time Management

PROJECTS

AI-Powered Gardening System

Python, Machine Learning, IoT Sensors

- Designed a Python-based irrigation system ingesting real-time soil moisture sensor data to predict optimal watering requirements.
- Trained and deployed an ML model to automate irrigation decisions, reducing manual intervention across watering cycles.
- Improved water usage efficiency by integrating data-driven predictions with hardware automation logic end-to-end.

F.L.A.M.E. – Fire Location And Mitigation Engine

Arduino Uno, Raspberry Pi, IoT, Python

- Developed an IoT-enabled autonomous fire detection and suppression system with real-time mobile monitoring interface.
- Integrated a 5-sensor hazard detection suite (smoke, toxic gases, flame, temperature, humidity) with automated pump activation for sub-second emergency response.
- Managed 5+ concurrent sensor data streams; designed a CNN-based roadmap for AI-driven visual fire detection and hazard classification.

CERTIFICATIONS & AWARDS

- **1st Prize Winner** – Python Coding Workshop organized by UEMK Computer Society of India (CSI); outperformed competing participants in algorithm design and implementation.
- **Vice Chancellor's Tech Maker Award** – Recognized for outstanding technical contribution and innovation by the University of Engineering & Management, Kolkata.